

Agricultural Futures Trading Decision Using AI Agent With Multiscale Candlestick Analysis

Jiarong Wang , Graduate Student Member, IEEE, Jiaji Wu , Member, IEEE, Guoshuai Zhang , Mingzhou Tan , Shaohong Chen , and Zeyi Lin 

Abstract—The high volatility, seasonality and complex trading environment of agricultural futures markets present significant challenges to the dynamic adaptability of algorithmic trading systems. Traditional methods rely on fixed-scale image analysis and lack adaptability, making them unsuitable for varying volatility conditions. Therefore, this study proposes an AI agent for agricultural futures trading decisions. First, an agricultural futures adaptive volatility rate serves as a computational tool to dynamically identify high-volatility regions and generate finer-scale candlestick charts. Second, the agent uses Vision-Language Large Model with robust image comprehension to analyze the multiscale candlestick charts and extract key features such as trend direction and technical patterns. Subsequently, a Large Language Model with advanced natural language understanding and logical reasoning serves as the “decision-making brain”, evaluating market trends and making buy/sell decisions. Finally, the agent refines its decision logic through a multimodal feedback mechanism that combines numerical and textual information from ongoing interactions with the environment, thereby enhancing system adaptability and robustness. Experimental results indicate that this framework significantly improves the accuracy, stability, and risk control of trading strategies, offering valuable insights for human decision-making. In addition, it demonstrates potential applicability in other financial markets such as stock indices, energy, and major commodities, providing an innovative solution for intelligent trading in complex market conditions.

Index Terms—AI agent, agricultural futures, algorithmic trading, multiscale candlestick analysis, vision-language large models.

I. INTRODUCTION

ARTIFICIAL intelligence (AI) technology has been widely used in financial activities such as stock markets, futures trading, and risk assessment. For example, in the U.S. stock exchange, about 75% of stock trades are executed by automated

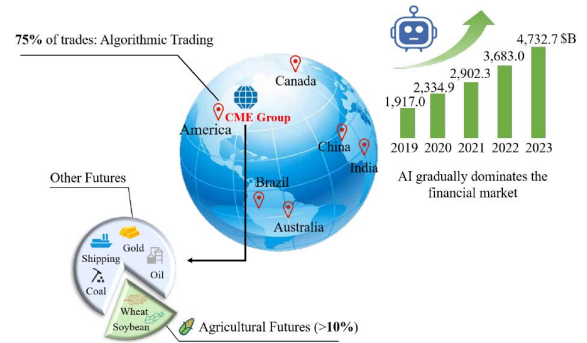


Fig. 1. Rising dominance of AI in global financial markets and the significant proportion of agricultural futures.

trading systems [1]. AI technology has significantly enhanced market stability and promoted technological advances in Social Fintech. The economic significance of agricultural futures has increased significantly after the 2006 food price crisis and the 2007–2008 financial crisis [2]. According to data from the CME Group in 2024, agricultural futures account for more than 10% of trading volume¹ (as shown in Fig. 1). However, compared with other financial markets, agricultural futures markets are more volatile [3]. This is mainly due to their seasonal characteristics, which show significant cyclic patterns [4]. This increases the complexity of algorithmic trading.

Traditional algorithmic trading (or quantitative trading) uses mathematical rules for automated decision-making [1]. These methods have limited performance when facing the nonlinear features and sudden volatility of agricultural futures markets [5]. Reinforcement learning combined with the feature extraction power of deep learning has significantly improved the adaptability of algorithmic trading [6]. In recent years, candlestick charts (K-line charts) have become an important tool for analysis and prediction in financial markets [7]. However, these methods use fixed multiscale image analysis strategies, lacking dynamic adjustment ability. Additionally, for agricultural futures, it is important to fully consider their seasonal characteristics to improve the applicability and accuracy of these methods.

With the increasing complexity of markets, AI Agents combined with visual and language large models demonstrate

Received 28 December 2024; revised 4 April 2025 and 25 June 2025; accepted 25 June 2025. Date of publication 16 July 2025; date of current version 3 December 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62271381; in part by China Postdoctoral Science Foundation under Grant 2024M752529; and in part by the Postdoctoral Fellowship Program of CPSF under Grant GZC20232057. (Corresponding author: Jiaji Wu.)

Jiarong Wang, Jiaji Wu, Guoshuai Zhang, Shaohong Chen, and Zeyi Lin are with the School of Electronic Engineering, Xidian University, Xi'an, Shaanxi 710071, China (e-mail: jiarongw@163.com; wuji@mail.xidian.edu.cn; zhangguoshuai@xidian.edu.cn; shaohong_chen@foxmail.com; 22021211836@stu.xidian.edu.cn).

Mingzhou Tan is with the AAMA Angel Investment Management (Beijing) Company Ltd., Beijing 100085, China (e-mail: tanmz@aamachina.com.cn).

Digital Object Identifier 10.1109/TCSS.2025.3583788

¹<https://www.cmegroup.com/market-data/browse-data/exchange-volume.html>

significant advantages in semantic understanding and task planning [8]. In recent years, AI Agents have been successfully applied in fields such as gaming [9] and data generation [10]. For example, NVIDIA's AI Blueprint uses vision-language large models (VLLM) and large language models (LLM) significantly improving video summarization performance [11]. Compared with traditional algorithmic trading, AI Agents offer autonomous learning and dynamic adjustment capabilities. Their ability to accurately interpret candlestick charts using VLM makes them particularly effective in addressing high volatility and seasonal characteristics in agricultural futures markets.

Therefore, this article proposes an AI agent-based trading framework following the "Robot Scientist" paradigm [12] and analyzes both overall trends and high-volatility segments in depth. It incorporates an agricultural futures adaptive volatility rate (AFAVR) and a VLLM-based multiscale candlestick chart analysis module to extract combined image features and volatility information. A multimodal feedback mechanism optimizes decision logic to generate precise and efficient trading strategies. This system significantly improves trading accuracy, reduces investment risks, and provides an innovative solution for Social Fintech, thereby fostering a trustworthy investment environment and a sustainable financial ecosystem.

Our contributions are as follows.

- 1) *AI Agent for Agricultural Futures Trading*: AFAVR identifies high-volatility regions and generates fine-scale candlestick charts from large-scale ones. VLLM analyzes multiscale candlestick features, and LLM generates trading decisions while optimizing decision logic with a multimodal feedback mechanism.
- 2) *AFAVR Calculation and Multiscale Visualization*: AFAVR uses a seasonally adaptive sliding window to generate volatility curves, identify high-volatility regions, and create fine-scale representations. The candlestick chart analysis tool leverages VLLM to capture market trends and volatility features, supporting precise trading decisions.
- 3) *Multimodal Feedback Mechanism*: Numerical feedback quantifies profit and loss for clear evaluation, while textual feedback records decision content and reasoning, enhancing interpretability and simulating human learning to improve decision logic.

The structure of this article is as follows: Section II reviews the research progress in futures trading systems. Section III introduces the proposed AI agent framework, integrating multiscale candlestick analysis, AFAVR, and a multimodal feedback mechanism to enhance decision-making. Section IV presents experiment and discussion, while Section V concludes with contributions and future research directions.

II. RELATED WORKS

A. Multimodal Data-Driven Futures Market Prediction and Analysis

Early research on futures market price prediction mainly relies on classical time series models, such as ETS [13] and ARIMA [14]. These methods have good interpretability and

stability but struggle to capture nonlinear features and sudden volatility due to their linear assumptions. To address these issues, researchers introduced machine learning and deep learning methods (e.g. BPNN [15] and ELM [16]) to enhance nonlinear feature extraction. With the diversification of data sources, research on integrating numerical data with text information from news and reports has emerged [17]. For example, Li et al. combined LSTM and SVR in the agricultural futures sector, incorporating news data [15].

Building on these studies, image-based information (such as candlestick charts) has become an important tool for futures market analysis due to its intuitive representation. Researchers have used deep learning models to automate feature extraction from candlestick charts [18]. Lin et al. proposed the candlestick trend and technical deviation (CTTD) algorithm, which tracks candlestick features using open-high-low-close (OHLC) data, achieving initial automation of feature extraction [19]. In recent years, researchers have combined deep reinforcement learning (DRL) techniques to optimize candlestick feature extraction and multiscale analysis processes [20]. For example, Fengqian and Chao combined K-line clustering analysis with RL to improve decision efficiency in feature spaces [21]. These methods demonstrate strong adaptability and optimization capabilities under complex market conditions.

Research on candlestick charts has evolved from single-pattern recognition to multiscale analysis, showing the potential of image feature mining in optimizing trading strategies. However, existing methods may weaken detailed features during multiscale overlay analysis. By introducing VLLM, which allow for finer and more interpretable feature mining of candlestick charts, the accuracy of futures market predictions can be further improved.

B. Algorithmic Trading Research

Traditional methods mainly rely on fundamental analysis and technical analysis but lacks quantitative standards for entry and exit timing [22]. The introduction of reinforcement learning (RL) overcomes the limitations of traditional methods by integrating prediction and decision-making. It continuously optimizes trading strategies to respond to dynamic changes in financial markets [23]. RL has shown significant robustness and adaptability in markets such as foreign exchange and stocks. For example, Rundo combined reinforcement learning with long short-term memory to enhance strategy flexibility and performance by deepening time-series structure and decision feedback [24]. Deep reinforcement learning (DRL) further improves traditional RL by combining deep neural networks for feature extraction [13]. In recent years, researchers have optimized DRL models through modular structures (such as parallel modules and LSTM), improving returns while effectively reducing volatility risk [25].

Due to its leverage effect, delivery time, and high volatility, the complexity of trading decisions in futures markets is higher. Current research still faces challenges in multimodal data fusion, trading constraint modeling, and market adaptability. A mature application framework has not yet been established.

C. Research on Agents in the Financial Domain

In the financial domain, early agents relied on fixed rules to simulate market behavior, which lacks flexibility and struggled to adapt to dynamic and complex market environments [26]. By enabling collaboration and division of labor among agents, multiagent systems (MAS) significantly improved market modeling and strategy adaptability [27]. When combined with DRL, MAS not only enhanced the robustness of trading strategies in response to market volatility [28] but also strengthened the dynamic behavior adjustment of agents in complex markets. For example, the adaptive agent model proposed by Kanzari and Said integrates fundamental analysis, technical analysis, and imitation behavior [29]. Agents have also been used to simulate the dynamic changes in market prices [30]. For instance, the study by Pérez-Pons et al. combined deep Q-learning with agricultural environmental indicators, achieving accurate predictions of futures prices and optimizing the related decision-making process [31].

With the rapid development of large models such as ChatGPT [32], the combination of agents and LLM has led to the concept of AI agents [33]. AI agents inherit the learning and decision-making advantages of traditional agents while leveraging the semantic understanding and multimodal processing capabilities of LLM. Introducing AI agents into the financial domain provides new directions and solutions for dynamic decision optimization.

Existing research on the futures market has made significant progress in numerical, text, and candlestick feature extraction. However, there are still limitations, particularly in the deep mining of image features, the application of large models, and the fusion of multiscale and multimodal data. This article aims to introduce VLM and AI agents, combining the extraction and analysis of multiscale features to further improve the stability and applicability of trading strategies. It seeks to provide more comprehensive and accurate solutions for futures trading decisions in complex market environments.

III. METHOD

A. Framework Design of the AI Agent for Agricultural Futures Trading Decisions

The overall framework of the agricultural futures market trading decision agent is shown in Fig. 2. In Fig. 2, module 1 and module 2 form the core tools of the agent, working together to complete the multiscale candlestick chart analysis. The AFAVR calculation module, as a key tool in the framework, identifies high-volatility areas in agricultural futures candlestick charts by calculating the AFAVR. These areas are then represented using candlestick charts with finer scales. This process continues until the details of each candlestick chart become clearly visible. The final multiscale candlestick charts comprehensively display market trends from global to local levels, as shown in the monthly, weekly, and daily charts in module 2 of Fig. 2.

Then, the powerful image recognition capabilities of the VLM, combine with precise and detailed prompts, are used to analyze the multiscale candlestick charts in-depth (module 2 in Fig. 2). This step functions as the agent's "eyes," extracting key trend features, important market points, and potential trading

signals from the candlestick charts. After completing the multiscale chart analysis, the agent proceeds to the buy/sell decision step (module 3 in Fig. 2). This step acts as the "brain" of the agent, using LLM and precise prompts (as shown in Appendix 3) to comprehensively assess the output from the analysis module. By considering market trends, volatility characteristics, and historical data, the decision module generates specific trading strategies, including buy, sell, or hold operations.

Finally, based on actual market price movements (up or down), the agent calculates the profit and loss resulting from the buy/sell decisions. The profit and loss calculation module in module 4 of Fig. 2 quantifies the market feedback after the decision is executed. The image analysis results, specific decisions, reasoning behind the decisions, and profit and loss outcomes are stored in the memory module as the agent's memory. In the next decision-making cycle, the agent can effectively combine numerical and textual information, optimizing the decision-making process and improving decision accuracy.

B. Tools Design: AFAVR Calculation and Multiscale Visualization

To analyze the agricultural futures market efficiently, two core tools were designed: AFAVR calculation and multiscale visualization. These tools quantify volatility and analyze market trends, providing accurate trend and volatility information for decision-making.

1) *Market Volatility Analysis Based on Agricultural Futures Adaptive Volatility Rate*: The price fluctuations in the agricultural futures market show strong seasonal characteristics. For example, during the sowing and harvest periods, factors such as weather and supply-demand influence cause price volatility to be significantly higher than at other times. As shown in Fig. 3(a), the green and red areas correspond to the sowing and harvest periods, where volatility is higher, while other periods exhibit relatively low market fluctuations. To better capture these characteristics, the AFAVR is proposed to identify periods with significant price volatility.

First, define the true range of volatility TR_t using (1) to measure the magnitude of market price fluctuations at time t . Select the maximum value from the following: the range between the highest and lowest prices $High_t - Low_t$, the range between the highest price and the previous closing price $|High_t - Close_{t-1}|$, and the range between the lowest price and the previous closing price $|Low_t - Close_{t-1}|$.

Secondly, the average true range ATR_t is calculated using (2). This represents the average of the actual price ranges TR_t over the past n_1 periods. The sliding window size n_1 determines the time frame for the market's average volatility level and can be adjusted flexibly based on market characteristics.

Finally, the adaptive futures volatility ratio $AFAVR_t$ is calculated using (4). This is determined by subtracting the average ATR [calculated using (3)] within the sliding window and dividing the result by ATR , converting it into a percentage. A positive value indicates stronger market volatility, while a negative value suggests weaker market volatility.

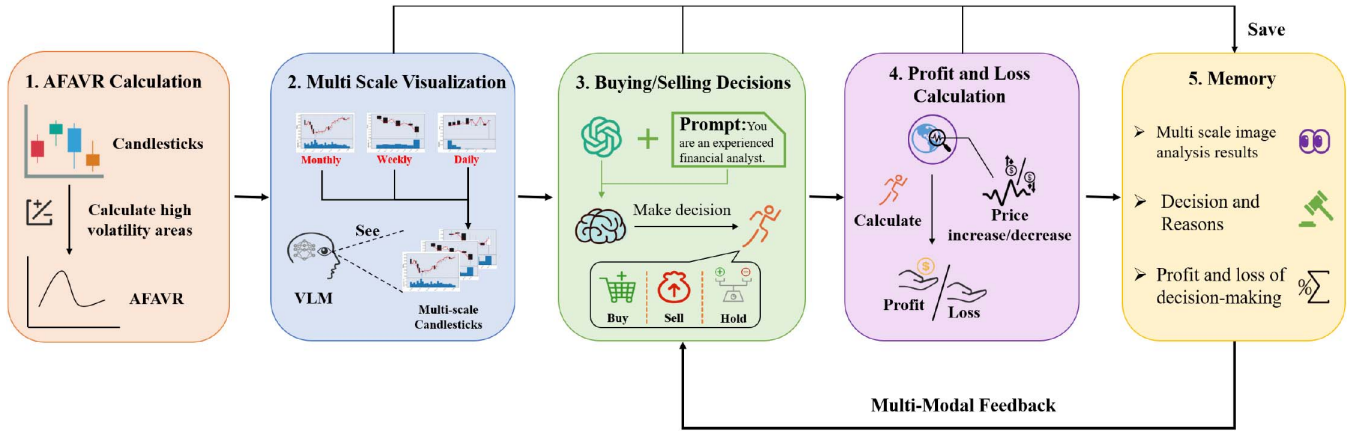


Fig. 2. Overall framework of the AI agent for agricultural futures trading decisions, consisting of five modules: 1. AFAVR calculation for identifying high-volatility areas; 2. Multiscale visualization for trend analysis using VLM; 3. Decision-making with LLM; 4. Profit and loss calculation based on market feedback; and 5. Memory storage for decision optimization.

As shown in Fig. 3, the AFAVR calculation is combined with the adaptive sliding window mechanism. During the sowing and harvest periods, when volatility is high, shorter sliding windows (e.g. $n = 2$) are used to capture rapid price changes. During other periods, longer sliding windows (e.g. $n = 3$) are used to reflect stable price trends. By connecting the AFAVR values of all time points, the AFAVR curve shown in Fig. 3(c) is obtained. Based on the preset threshold, this method identifies high-volatility periods and evaluates the market volatility intensity within these periods

$$TR_t = \max \left(\text{High}_t - \text{Low}_t, \right. \\ \left. |\text{High}_t - \text{Close}_{t-1}|, \right. \\ \left. |\text{Low}_t - \text{Close}_{t-1}| \right) \quad (1)$$

$$ATR_t = \frac{1}{n_1} \sum_{i=0}^{n_1-1} TR_{t-i} \quad (2)$$

$$\overline{ATR} = \frac{1}{n_2} \sum_{i=1}^{n_2} ATR_{t-i} \quad (3)$$

$$\text{AFAVR}_t = \frac{ATR_t - \overline{ATR}}{\overline{ATR}} \times 100\%. \quad (4)$$

2) Multiscale Candlestick Analysis Based on Vision-Language Large Models: As shown in Fig. 3(a), the framework first analyzes the overall market trend using a larger time scale (e.g. monthly chart), capturing long-term characteristics. Based on this analysis, AFAVR is applied to preliminarily screen for high-volatility regions, identifying key areas for further detailed analysis. For instance, in Fig. 3(c), the AFAVR curve for April and May is significantly above the threshold, providing a basis for further refined analysis of these target regions.

This screening process is not mechanical but adaptive, adjusting based on market volatility characteristics. Regions with low volatility are analyzed only at larger time scales, while regions with high volatility dynamically switch to smaller time scales (e.g. weekly or daily charts) for more detailed information. As

shown in Fig. 3(d), the weekly chart provides a zoomed-in analysis of high-volatility areas, illustrating a significant upward price trend in April and May.

Finally, as illustrated in Fig. 3(b), the VLM's advanced image recognition capabilities are used, combined with precise prompt settings (as detailed in Appendix 2), to extract feature information across different scales of candlestick charts.

C. Multimodal Feedback Optimization Mechanism

After the agent executes a buy or sell operation, the system calculates the trading outcome (e.g. profit or loss) based on actual market price movements. This module evaluates the effectiveness of the agent's decisions in a quantitative manner. For instance, in Fig. 4, the decision to "buy 50 contracts" results in a profit of \$100. Numerical feedback provides the agent with immediate references for strategy adjustment and helps it balance risks and rewards more accurately.

In parallel with numerical evaluations, textual feedback plays a key role in enhancing the interpretability and adaptability of the agent's decision-making process. This feedback captures short-term historical trading information stored in the memory module, including market trends, executed decisions, decision rationales, and resulting outcomes. As shown in Fig. 4, this information enables the agent to retrospectively examine complete decision cycles—for example, identifying that a decision to "Buy 50 contracts" was based on a prior trend assessment ("The price is expected to rise") and supported by specific reasoning. By explicitly linking past decisions with their underlying logic and market responses, textual feedback provides a transparent reference for the agent to learn from its actions and refine future strategies accordingly.

The multimodal feedback mechanism integrates numerical and textual feedback through the memory module, serving as a core support for optimizing the agent's decision-making. By combining historical data and real-time feedback, the agent can dynamically adjust its strategy, providing more comprehensive and accurate support for the next round of decisions.

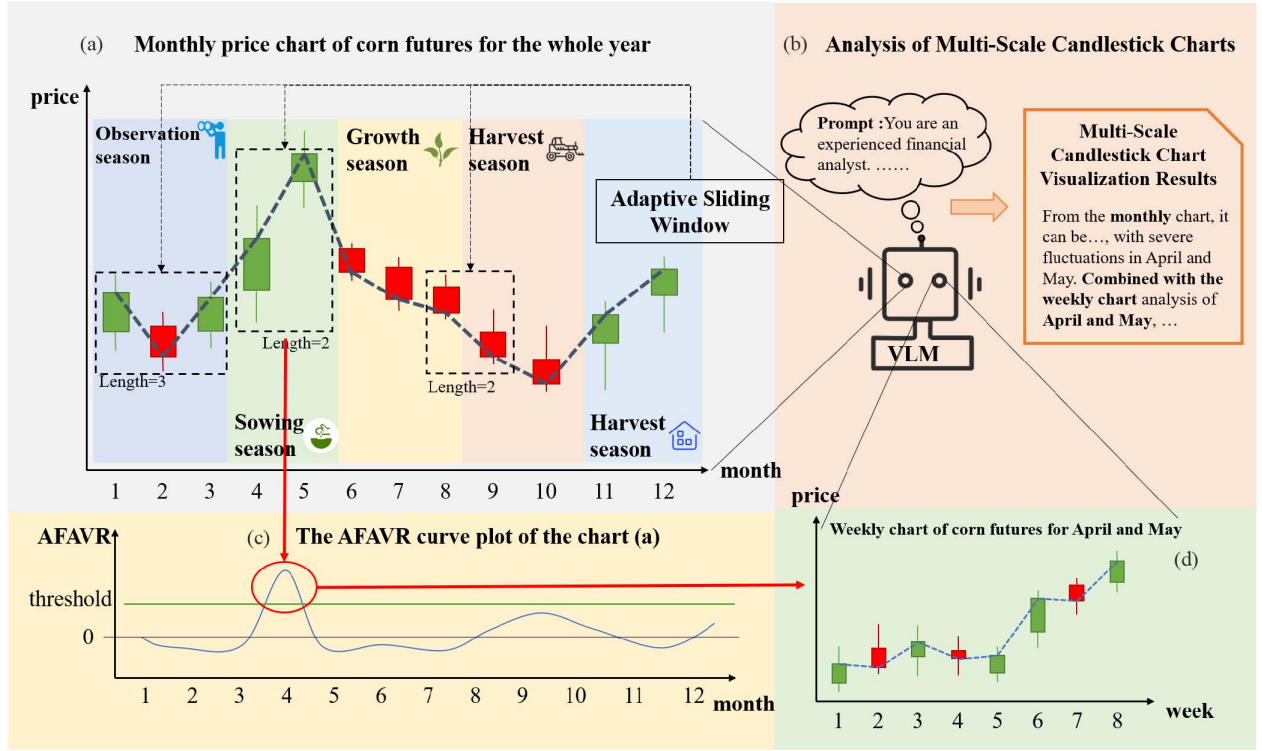


Fig. 3. AFAVR calculation based on the adaptive sliding window mechanism and multiscale adaptive analysis using candlestick charts. (a) AFAVR calculation on a monthly chart with an adaptive sliding window mechanism set according to seasonal characteristics. (b) Identifying volatility peaks using the AFAVR curve and a preset threshold. (c) Detailed analysis of high-volatility months using weekly charts. (d) AI agent derives visualization results by combining VLM with multiscale analysis.

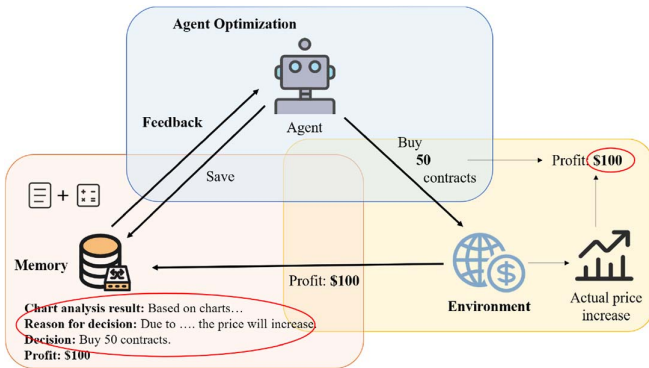


Fig. 4. Multimodal feedback optimization mechanism combining numerical and textual feedback. The agent generates trading decisions (e.g., buying 50 contracts) based on chart analysis results and calculates actual profits (e.g., \$100) through environmental feedback. These outcomes, along with decision rationales and profit/loss information, are stored in the memory module to support decision optimization.

IV. EXPERIMENT

A. Experimental Setup

To more accurately simulate real trading conditions, this experiment begins from a zero-position state—i.e., with no initial holdings—and determines the initial buy or sell action based on the agent's first assessment of market trends. This approach aligns with the practical decision-making process of entering a market and enhances the authenticity of the trading scenario. In

addition, we adopt a $T + 1$ trading mechanism, whereby profits and losses from a given day's trading decisions are realized on the following day. This structure assists in evaluating the agent's ability to adapt to short-term market fluctuations. The LLM employed in the experiment is GPT-4o, with the *max_tokens* parameter set to 10 000 to ensure the generation of sufficiently comprehensive content.

B. Dataset

To validate the effectiveness of the proposed AI agent, this study conducts an empirical analysis using two representative agricultural futures products: Chinese soybean futures and U.S. wheat futures. Soybean futures, traded on the Dalian commodity exchange (DCE), have a long history and high trading volume [34], making them a major agricultural futures product globally [35]. U.S. wheat futures are also highly liquid and play a critical role in the global food supply chain [36]. This study uses data from the a2309 soybean futures contract (settled in September 2023) traded on the DCE,² and 2022 wheat futures data from Kaggle.³

To capture the seasonal changes and market volatility characteristics of agricultural futures, the dataset includes one year of trading records. For data division, the first 10 months are used to

²<http://www.dce.com.cn/>

³<https://www.kaggle.com/datasets/lusfernandotorres/us-wheat-futures-zw/data>

calculate AFAVR and analyze market trends. The last 2 months are used to test the decision-making ability of the AI Agent. Specifically, since the soybean a2309 contract was settled in September, data from January to July are used for analysis, and data from August and September are used for validation. For wheat futures, January to October are used as analysis data, and November and December as validation data. The selection of data from multiple markets and regions helps comprehensively assess the applicability and effectiveness of the framework.

C. Evaluation Metrics

To comprehensively assess the performance of the agent in futures trading, we employ multiple commonly used evaluation metrics in this field that quantify aspects of profitability, risk control, and decision stability. The specific metrics and their calculation methods are as follows.

- 1) *Cumulative Return Curve*: This metric records the changes in cumulative returns throughout the experiment. It reflects overall strategy performance and long-term profitability.
- 2) *Single Trade Return Curve*: This metric records the return of each individual trade. It aids in evaluating the agent's performance and risk control at the single-trade level.
- 3) *Average Trade Return*: This metric measures the agent's average profitability per trade. A higher average trade return indicates that the agent consistently achieves greater profits on individual trades, reflecting superior decision-making quality. The calculation formula is as follows:

$$\text{Average Trade Return} = \frac{\text{Total Profit}}{\text{Number of Trades}}. \quad (5)$$

- 4) *Win Rate*: This metric measures the probability that the agent makes a correct decision. A higher win rate suggests that the agent correctly judges market conditions in most cases. The calculation formula is as follows:

$$\text{Win Rate} = \frac{\text{Number of Winning Trades}}{\text{Total Number of Trades}} \times 100\%. \quad (6)$$

- 5) *Profit Factor*: This metric assesses the balance between the agent's profitability and its risk of loss. A profit factor greater than 1 indicates that the strategy is profitable overall. The profit factor is a key indicator of strategy stability and risk management. Its calculation formula is as follows:

$$\text{Profit Factor} = \frac{\text{Total Profits}}{\text{Total Loss}}. \quad (7)$$

- 6) *Maximum Drawdown (MDD)*: This metric measures the largest peak-to-trough decline in the agent's cumulative returns. A smaller MDD indicates better stability and risk management. The calculation formula is as follows:

$$\text{MDD} = \max(\text{Peak Value} - \text{Trough Value}). \quad (8)$$

- 7) *Volatility*: This metric quantifies the degree of fluctuation in the agent's single-trade returns, typically measured as the standard deviation of per-trade returns. Lower

TABLE I
AFAVR EXPERIMENTAL PARAMETER SETTINGS

Scale	Adaptive Sliding Window		Fixed Sliding Window	Threshold
	High-Volatility Period	Other Periods		
Monthly	2	4	4	10
Weekly	3	5	5	

volatility indicates more stable returns and lower risk. The calculation formula is as follows:

$$\text{Volatility} = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (\text{Profit}_i - \overline{\text{Profit}})^2} \quad (9)$$

where N is the number of trades, Profit_i is the return of i trade, and $\overline{\text{Profit}}$ is the average return per trade.

- 8) *Annualized Sharpe Ratio*: This metric measures the agent's returns relative to the risk undertaken, annualized for comparison. The annualized Sharpe ratio reflects the quality of the agent's returns on a risk-adjusted basis. A higher Sharpe ratio indicates greater returns per unit of risk. The calculation formula is as follows:

$$\text{Sharpe Ratio} = \frac{\text{Mean Daily Profit}}{\text{Standard Deviation of Daily Profit} \times \sqrt{\text{Annual Trading Days}}}. \quad (10)$$

D. AFAVR's Ability to Extract High-Volatility Regions

This experiment aims to validate the accuracy of the agricultural futures adaptive volatility Rate (AFAVR) in identifying high-volatility regions and to explore the effectiveness of adaptive sliding windows in agricultural futures markets. The experiment has two primary objectives: 1) to confirm that AFAVR can accurately identify periods of high market volatility and provide detailed insights through progressively zoomed candlestick charts; and 2) to verify that adaptive sliding windows can effectively identify key periods with seasonal volatility characteristics, such as planting and harvest seasons, in agricultural futures.

To achieve these objectives, two sets of comparative experiments were designed, using soybean futures as an example: one group used adaptive sliding windows, while the other used fixed sliding windows for volatility calculation. The experimental parameters are shown in Table I.

The experimental results are shown in Figs. 5 and 6. In the adaptive sliding window group, AFAVR accurately identified periods of high volatility. As shown in Fig. 5(a), the AFAVR values for soybean futures contract significantly exceeded the threshold during 4.10–5.20 and 6.10–7.10, indicating high market volatility during these periods. However, the monthly chart output showed high-volatility regions as 3.31–6.4 and 5.31–7.16 [Fig. 5(b) and (c)]. This difference is attributed to the monthly chart's calculation, which uses monthly averages, causing end-of-month dates to represent the month's overall volatility and smoothing intraday fluctuations, thereby reducing noise. This ability to accurately identify high-volatility periods was further validated on the weekly chart. For instance, during 3.31–6.4, while the monthly chart indicated an overall downward trend, the weekly chart revealed detailed price

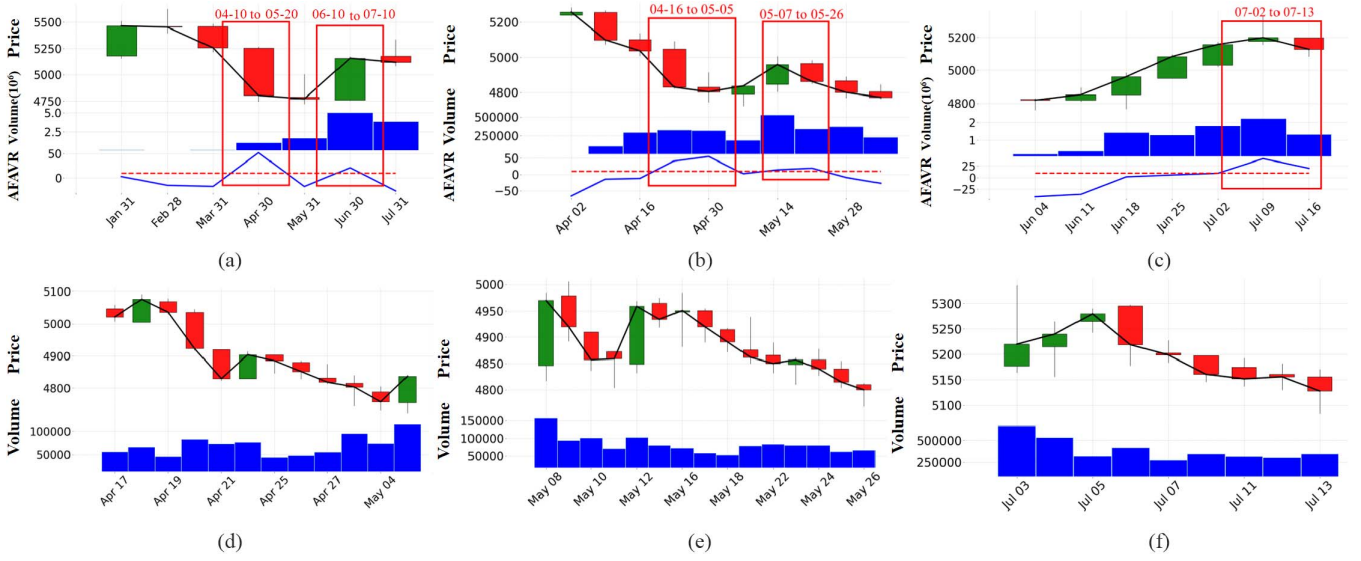


Fig. 5. AFAVR results of soybean futures with adaptive sliding window. (a) Monthly chart of soybean futures, with red boxes indicating high-volatility regions. (b) Weekly charts of the red-boxed regions (03-31 to 06-04) in (a), where the red boxes represent high-volatility areas at the weekly scale. (c) Weekly charts of the red-boxed regions (05-31 to 07-16) in (a), where the red boxes represent high-volatility areas at the weekly scale. (d) Daily charts of the high-volatility regions (04-16 to 05-05) are identified in (b). (e) Daily charts of the high-volatility regions (05-07 to 05-26) are identified in (b). (f) Daily charts of the high-volatility regions (07-02 to 07-13) are identified in (c).

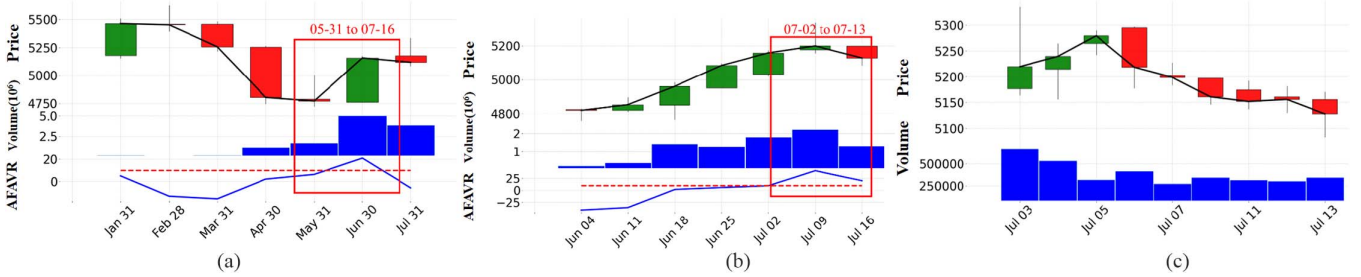


Fig. 6. AFAVR results of soybean futures with fixed sliding window. (a) Monthly chart of soybean futures, with red boxes marking high-volatility regions under the fixed sliding window. (b) Weekly chart of the red-boxed region in (a), where the red boxes indicate high-volatility areas at the weekly scale. (c) Daily chart of the high-volatility region highlighted in the weekly chart.

fluctuations during this period, including phases of recovery [Fig. 5(b)].

To capture more details, the high-volatility regions in the weekly chart were further zoomed into daily charts for AFAVR calculation [Fig. 5(d)–(f)]. The results demonstrated that the adaptive sliding window effectively narrowed the high-volatility regions step-by-step and accurately displayed subtle market price changes, offering a multi-scale perspective on market volatility analysis.

In contrast, the fixed sliding window group's results showed significant limitations. As seen in Fig. 6(a), the monthly chart only identified the high-volatility region from 5.31–7.16, failing to detect the 3.31–6.4 period. Further analysis using weekly and daily charts also reflected the lower precision of the fixed sliding window in identifying high-volatility regions, missing key periods of significant volatility. Specifically, during 3.31–6.4, the fixed sliding window failed to capture notable market recovery details, leading to a lack of critical information in subsequent weekly and daily charts.

In summary, the experimental results demonstrate that AFAVR has high accuracy in extracting high-volatility regions in agricultural futures markets. Compared with the fixed sliding window method, dynamically adjusting the window size based on seasonal patterns effectively avoids missing key high-volatility regions.

E. Effectiveness of Multiscale Candlestick Analysis Based on Vision-Language Large Models

To verify the effectiveness of multiscale adaptive analysis framework, we conducted three comparative experiments. Our goal was to confirm that the proposed method can effectively magnify image details and improve the understanding of future price trends.

- 1) Directly input the original soybean daily futures chart (Fig. 7) into the large model without any processing.
- 2) The VLM receives the multiscale adaptive analysis results derived from AFAVR-based decomposition (as shown in Fig. 5).

TABLE II
ANALYSIS OF MULTISCALE ADAPTIVE ANALYSIS FRAMEWORK EXPERIMENTAL RESULTS

Category	Group 1 Results	Group 2 Results	Group 3 Results	Winner
Price Trend	<i>Pros:</i> Reflects mid- to long-term trends well, shows monthly-level long-term changes. <i>Cons:</i> Fails to capture key fluctuations in April–May, lacks short-term trend analysis.	<i>Pros:</i> Accurately identify April–May and June–July fluctuation periods with clear details. <i>Cons:</i> Slight shortfall in linking multiple scales smoothly.	<i>Pros:</i> Shows overall trend, identifies main volatile areas. <i>Cons:</i> Some details ignored, not fully capturing short-term fluctuations.	Group 2
Key Levels	<i>Pros:</i> Some details are clear; overall highlights key fluctuations moderately. <i>Cons:</i> Some detailed fluctuations not covered; short-term characteristics not fully revealed.	<i>Pros:</i> Accurately identifies key levels at 4800 and 5200, showing clear volatility patterns and rich detail. <i>Cons:</i> High-resolution analysis requires larger display space.	<i>Pros:</i> Identifies 4800 support and 5300 resistance, relatively stable. <i>Cons:</i> Identification of 4800 support slightly less refined than 5200.	Groups 2 & 3
Technical Patterns	<i>Pros:</i> Identifies head-and-shoulders, ascending wedges, predicting short-term price shifts. <i>Cons:</i> Lacks fine-grained depiction of subtle short-term rebounds and pressures.	<i>Pros:</i> Multi-scale integration fully recognizes patterns, especially double tops and bottoms. <i>Cons:</i> Some marginal pattern features need re-checking.	<i>Pros:</i> Multi-scale visualization reveals double-top patterns, helping anticipate market downturns. <i>Cons:</i> Short-term predictive accuracy for special patterns needs improvement.	Groups 2 & 1
Volume Analysis	<i>Pros:</i> Focuses on June–July volatility, offering insights for event-rich periods. <i>Cons:</i> Weak correlation between volume and price changes; limited analytical precision.	<i>Pros:</i> Closer matching of volume with price volatility, showing more detailed trends. <i>Cons:</i> Some details need more depth; short-term volatility not fully explored.	<i>Pros:</i> Volume elements are more intuitive and clearer than the original chart. <i>Cons:</i> Not fully capturing abrupt short-term volume spikes.	Group 2
Seasonality/Periodicity	<i>Pros:</i> Reflects June–July seasonal volatility, reveals May–July summer market traits. <i>Cons:</i> Does not fully capture year-round cyclical features.	<i>Pros:</i> Accurately recognizes seasonal volatility features (e.g., May–July, Jan–Feb) with comprehensive multi-scale analysis. <i>Cons:</i> Interplay between seasonality and other fluctuations needs further exploration.	<i>Pros:</i> Clear June–July cyclical traits, useful for policy forecasts in specific periods. <i>Cons:</i> Not adequately adaptive to full-year patterns.	Groups 2 & 1
Risk Evaluation	<i>Pros:</i> Clearly indicates support and resistance levels, helpful for short-term risk management. <i>Cons:</i> Does not precisely control fluctuation pace; somewhat rough risk assessment.	<i>Pros:</i> Time-precision assists risk judgment, providing accurate targets for strategy-making. <i>Cons:</i> Slightly insufficient fine-tuning in some periods; risk control could be more precise.	<i>Pros:</i> High time resolution aids in identifying risk points. <i>Cons:</i> Needs more refined detail; somewhat lacking precise risk control measures.	Groups 2 & 3
Multiscale Consistency	<i>Pros:</i> Shows basic connections between long-term and short-term trends. <i>Cons:</i> Lacks detailed depiction of short- and mid-term volatility.	<i>Pros:</i> Emphasizes deep linkage between short- and long-term, aiding insight into subtle market fluctuations. <i>Cons:</i> Some information redundancy.	<i>Pros:</i> Enhances correlation between long-term trends and short-term fluctuations, aiding local trend interpretation. <i>Cons:</i> Insufficient information.	Group 2 & 3

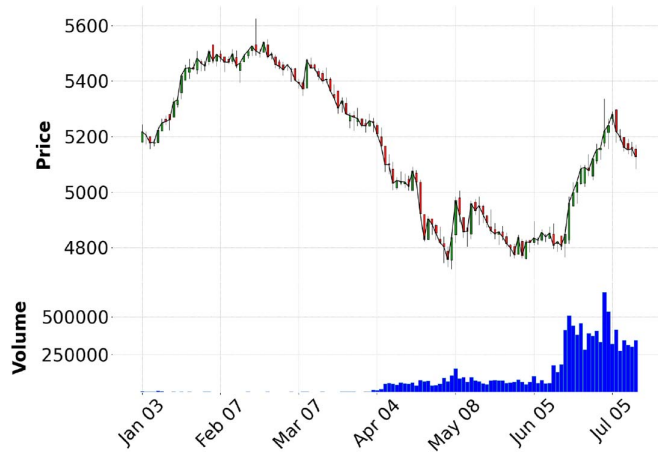


Fig. 7. Soybean futures daily candlestick chart (January–July).

3) The VLM receives the multiscale adaptive analysis results derived from a fixed-window volatility-based decomposition (as shown in Fig. 6).

Due to space limitations, the analysis results are presented in the Appendix 1.

The experimental results are shown in Table II, the three columns represent the outcomes of each experiment group respectively, and the final column (“winner”) highlights the group that demonstrated superior performance in each evaluation dimension. The VLLM evaluated the inputs from seven perspectives: price trend, key levels, technical patterns, volume analysis, seasonality/periodicity, risk evaluation, and multiscale consistency, listing the advantages and limitations of each method across these criteria.

Compared with the first and third experiments, the second experiment performs better in detecting price trends, support and resistance levels, technical patterns, volume-related changes, volatility evaluation, and trend continuity. It excels at revealing fine details and microlevel fluctuations in highly volatile regions. This advantage enables it to more precisely uncover the underlying patterns of price changes, providing a reliable basis for market forecasting. Therefore, the multiscale analysis method used in the second experiment has high practical value

TABLE III
EVALUATION METRICS FOR AGENT TRADING EXPERIMENTS UNDER NONFEEDBACK AND FEEDBACK MECHANISMS

Class	Evaluation	without_feedback		feedback	
		Daily	Weekly	Daily	Weekly
Soybean	Average trade return	384	1042	718.89	3868.57
	Win rate	71.11%	57.14%	88.89%	85.71%
	Profit factor	1.95	1.60	6.73	7.51
	MDD	9000.00	12230.00	3660.00	4160.00
	Volatility	2072.15	6138.71	1191.34	5881.30
	Annualized Sharpe ratio	3.63	2.70	9.58	10.44
Wheat	Average trade return	332.19	206.07	415.88	478.57
	Win rate	75.00%	71.00%	86.00%	86.00%
	Profit factor	3.53	1.49	8.94	5.21
	MDD	1737.50	2925.00	840.00	795.00
	Volatility	779.33	1431.20	657.15	771.67
	Annualized Sharpe ratio	6.77	2.29	10.05	9.84

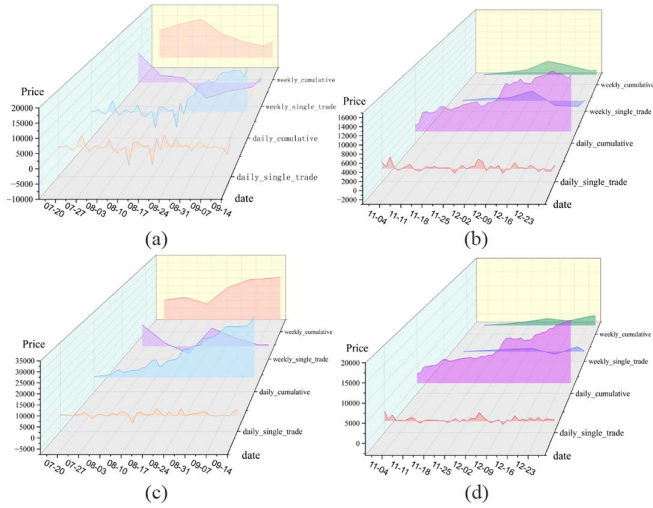


Fig. 8. Trading decision results of soybean and wheat futures under non-feedback and feedback mechanisms. (a) Weekly and daily single-trade and cumulative profit curves for soybean futures without feedback. (b) Weekly and daily single-trade and cumulative profit curves for wheat futures without feedback. (c) Weekly and daily single-trade and cumulative profit curves for soybean futures with feedback. (d) Weekly and daily single-trade and cumulative profit curves for wheat futures with feedback.

and shows strong potential for broader application in the futures market.

F. Accuracy of AI Agent Trading Decisions

This experiment analyzed the decision-making ability of the AI Agent for soybean and wheat futures under both no-feedback and feedback mechanisms. The goal was to evaluate the performance of the agent across daily and weekly time scales. Results indicate that under the no-feedback mechanism, both daily and weekly strategies showed good adaptability within their respective time scales. However, the feedback mechanism significantly improved decision accuracy, flexibility, and

stability. The results and corresponding evaluation metrics are presented in Fig. 8 and Table III.

Under the no-feedback mechanism, the agent demonstrated different adaptability to short-term fluctuations and medium-to-long-term trends. As shown in Fig. 8(a), for soybean futures, the daily strategy successfully captured the upward trend and built positions when the market broke out of the consolidation zone from mid-August to early September (as shown in Fig. 9). This was supported by increased trading volume, resulting in a rapid increase in cumulative profit and a win rate of 71.11%. In contrast, the weekly strategy focused on clear trend phases, such as late July to mid-August, where it filtered out short-term noise and achieved higher single-trade profits (1042 compared with 384). However, its lower sensitivity to price reversals led to larger drawdowns (12230.00) and higher volatility (6138.71), with an annualized Sharpe ratio of 2.70, which was lower than the daily strategy's 3.63. Similarly, Fig. 8(b) shows that for wheat futures, the daily strategy responded promptly to technical signals at key support levels during the November downturn (as seen in Fig. 10) and established long positions in December as the market rebounded above resistance levels, maintaining stable profitability with a win rate of 75.00% and an average single-trade profit of 332.19. Meanwhile, the weekly strategy effectively captured primary trends during the initial rebound phase, filtering out short-term noise. However, its slower adjustments during the volatile phase in November resulted in higher volatility and a lower annualized Sharpe ratio (2.29 compared with 6.77). Despite this, the weekly strategy still showed strong single-trade profitability in sustained trending markets.

With the introduction of the feedback mechanism, the agent showed significant improvements in trend recognition and risk control. As shown in Fig. 8(c) and (d), daily strategies achieved smoother cumulative profit curves by quickly adjusting positions and reducing drawdowns. For example, in soybean futures, the maximum drawdown for the daily strategy decreased from 9000.00 (no-feedback) to 3660.00 (feedback), while volatility dropped to 1191.34. The weekly strategy, on

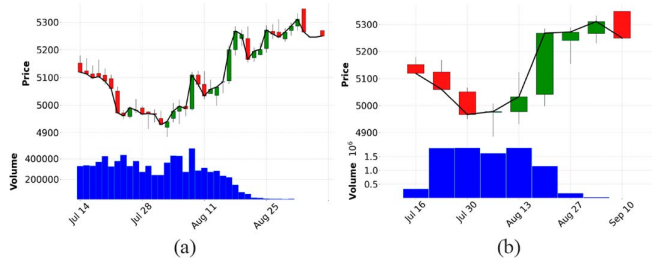


Fig. 9. Actual daily and weekly candlestick charts of soybean futures for the last 2 months of a year. (a) Soybean daily candlestick. (b) Soybean weekly candlestick.

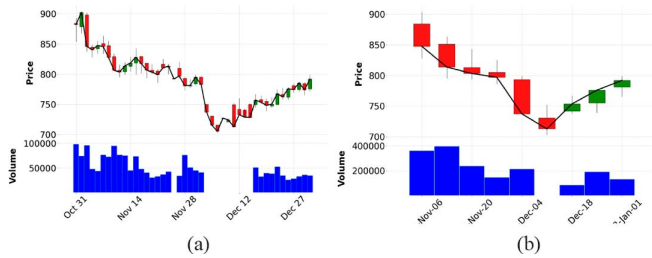


Fig. 10. Actual daily and weekly candlestick charts of wheat futures for the last 2 months of a year. (a) Wheat daily candlestick. (b) Wheat weekly candlestick.

the other hand, maintained steady growth in cumulative profits due to higher single-trade profitability, with an average single-trade profit of 3868.57. In terms of risk control, the feedback mechanism enhanced the agent's ability to adjust during high-volatility phases. Although the weekly strategy experienced greater volatility risks, it achieved a better balance between risk and return. For instance, the annualized Sharpe ratio for the weekly strategy in soybean futures improved to 10.44, surpassing the daily strategy's 9.58, while for wheat futures, both strategies showed similar Sharpe ratios (9.84 compared with 10.05).

In summary, the AI agent with feedback mechanisms achieved dual improvements in profitability and risk control. The daily strategy relied on a high win rate and flexible position adjustment to maintain lower drawdowns and volatility, while the weekly strategy demonstrated stronger profitability in trending markets due to higher single-trade gains.

G. Discussion

Experiments IV-D and IV-E clearly demonstrate the critical role of multi-scale analysis, combined with the AFAVR, in capturing market volatility. By dynamically selecting scales from larger (e.g., monthly) to smaller (e.g., daily), the framework effectively identifies high-volatility regions and focuses its analysis, avoiding the redundancy or omissions associated with fixed-scale methods. Specifically, in trending markets (e.g., the monthly chart in Fig. 5) and volatile markets (e.g., the weekly chart in Fig. 5), multiscale analysis highlights the complementary relationships between different time scales, significantly improving the accuracy of volatility detection.

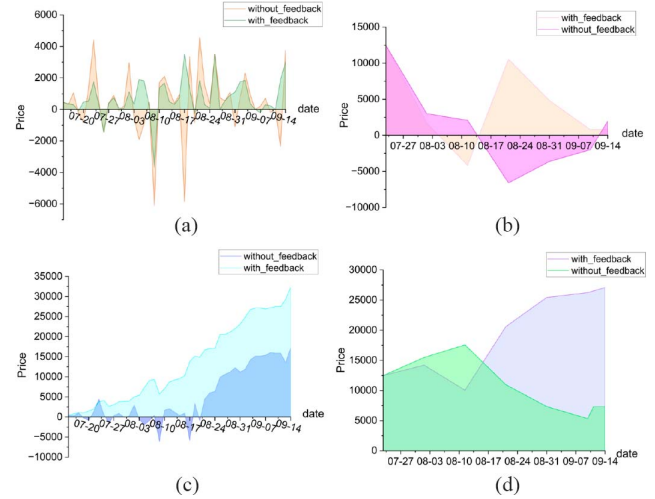


Fig. 11. Soybean multiscale profit curve with feedback. (a) Soybean daily single-trade profit curve. (b) Soybean weekly single-trade profit curve. (c) Soybean daily cumulative profit curve. (d) Soybean weekly cumulative profit curve.

By further integrating a VLLM, the proposed intelligent framework enables comprehensive understanding of market dynamics through in-depth analysis of multi-scale candlestick charts. VLLM demonstrate exceptional capability in extracting key features from candlestick charts, such as trend direction, critical support and resistance levels, while presenting the results clearly in structured textual form. For example, in Experiment IV-E, the daily chart successfully identified multiple key support and resistance breakouts during a high-volatility period in early August, providing clear guidance for subsequent trading strategies. This feature extraction capability addresses the limitations of traditional algorithms in multi-scale analysis and provides a reliable foundation for AI agents.

Building on this, a feedback mechanism was introduced to further optimize the agent's dynamic decision-making ability. The feedback mechanism adjusts the agent's behavior in real-time based on trading results, enabling rapid correction of erroneous signals and significantly reducing delays and cumulative errors. Figs. 11 and 12 compare the experimental results under feedback and no-feedback conditions on a unified scale, highlighting the notable differences between the two strategies.

From the daily single-trade return curves [as shown in Figs. 11(a) and 12(a)], strategies with feedback consistently outperformed those without feedback. This trend was observed in both soybean and wheat markets, where feedback strategies exhibited faster and more accurate adaptation to short-term fluctuations. For soybean futures, during the volatile period from late July to early August, the daily single-trade return curve under no-feedback conditions showed frequent negative values, whereas the feedback strategy demonstrated rapid "self-correction," significantly reducing both the frequency and duration of losses. Similarly, in wheat futures, during the volatile period from November to December, the feedback strategy showed smoother daily return fluctuations, markedly reducing the risk of consecutive losses. Similar patterns were also observed in the weekly return curves [as shown in Figs. 11(b)

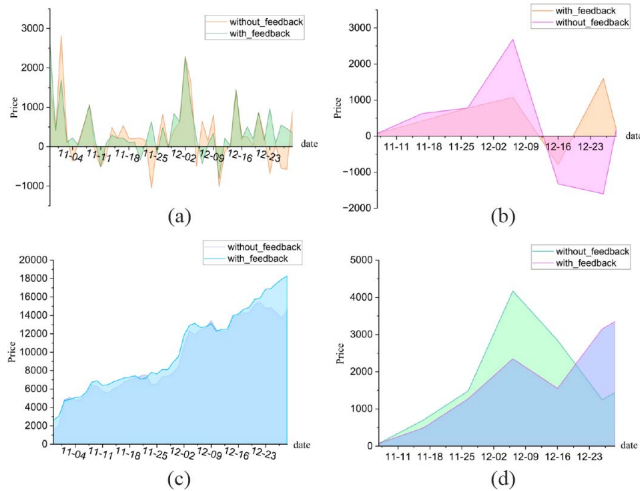


Fig. 12. Wheat multiscale profit curve with feedback. (a) Wheat daily single-trade profit curve. (b) Wheat weekly single-trade profit curve. (c) Wheat daily cumulative profit curve. (d) Wheat weekly cumulative profit curve.

and 12(b)], further validating the effectiveness of the feedback mechanism across different time scales.

This difference is also evident in the cumulative return curves. Feedback strategies achieved more stable and consistent growth for both soybean and wheat futures [as shown in Fig. 11(c), (d) and Fig. 12(c), (d)], indicating that the agent could adjust positions or close trades in short cycles to prevent sustained losses. While no-feedback weekly strategies often maintained positive returns, they remained insufficient in responding to sudden short-term countertrends.

The feedback mechanism empowers the AI agent with dynamic adjustment and self-correction capabilities. Moreover, the integration of high-quality multiscale image analysis and VLLM interpretation provides a complete solution for intelligent trading in complex markets. However, with the widespread application of large models, trading agents may face potential risks from “model hallucinations,” where overly confident predictions may deviate from actual data in specific niche markets or special scenarios. Addressing this issue requires introducing effective constraints and supervision on model outputs within the feedback framework. Establishing robust risk warning and adaptive correction mechanisms at the strategy execution level will be an important direction for future research.

Distinct from traditional postanalysis tools, the proposed feedback mechanism is embedded directly within the decision loop. After each trading action, the agent immediately receives numerical feedback (e.g., profit/loss) and textual feedback (e.g., decision rationale and market behavior), which are stored and directly influence the next round of strategy generation. This looped structure ensures that feedback is timely and relevant, enabling the agent to adapt quickly to changing market dynamics. The real-time incorporation of feedback, as demonstrated in Section IV-F, plays a vital role in enhancing decision quality across trading cycles.

Although the framework integrates large-scale models, the entire decision-making loop completes within 1–3 s, making it feasible for deployment in intraday or low-frequency trading

systems. In future work, we aim to explore model compression, distillation, and hardware-aware prompt optimization to further reduce latency and enhance scalability, especially for resource-constrained or edge deployment scenarios.

V. CONCLUSION

This study proposes an AI agent to address the challenges of high volatility and strong seasonality in agricultural futures markets. AFAVR utilizes a seasonally adaptive sliding window to dynamically identify high-volatility regions, enabling precise analysis of critical time periods and overcoming the limitations of fixed-scale methods. VLLM enhances system capabilities by extracting key features from multiscale candlestick charts, including trend directions, critical support and resistance levels, and technical patterns. Combined with a multimodal feedback mechanism that incorporates profitability evaluations and decision logs, the framework improves error correction and establishes a closed-loop optimization process, effectively adapting to dynamic market environments. Experiments on wheat and soybean futures demonstrate the framework’s exceptional performance in enhancing trading outcomes. The integration of multiscale analysis and feedback mechanisms significantly improves the precision of trend and volatility detection. Feedback-enabled strategies achieve 10%–30% higher prediction accuracy compared with no-feedback strategies and consistently outperform in reducing drawdowns and increasing returns. Beyond agricultural futures, the modular design of our framework makes it highly adaptable to other financial markets, including stock indices, energy futures, metal commodities, and even stock markets. The core components—such as AFAVR-based volatility detection, multiscale candlestick analysis, VLLM/LLM-driven decision-making, and multimodal feedback—can be customized or extended with domain-specific features for different asset classes. This demonstrates the framework’s potential as a scalable, robust, and interpretable solution for intelligent trading across a wide range of market environments. Nevertheless, challenges such as high computational costs and the risks of model hallucinations remain to be addressed in future research.

REFERENCES

- [1] E. P. Chan, *Quantitative Trading: How to Build Your Own Algorithmic Trading Business*, Hoboken, NJ: John Wiley & Sons, 2021.
- [2] M. M. Kupabado and J. Kaehler, “Price effects of commodity financialization: Review of the evidence,” *J. Econ. Surv.*, vol. 39, no. 1, pp. 352–374, Feb. 2025.
- [3] M. Massahi and M. Mahootchi, “A deep q-learning based algorithmic trading system for commodity futures markets,” *Expert Syst. Appl.*, vol. 237, 2024, Art. no. 121711.
- [4] Y. Li, Q. Liu, D. Miao, and Y. Tse, “Return seasonality in commodity futures,” *Int. Rev. Econ. Finance*, vol. 93, pt. B, pp. 448–462, Jun. 2024.
- [5] A. Spelta, N. Pecora, and P. Pagnottoni, “Chaos based portfolio selection: A nonlinear dynamics approach,” *Expert Syst. Appl.*, vol. 188, 2022, Art. no. 116055.
- [6] Z. Huang, N. Li, W. Mei, and W. Gong, “Algorithmic trading using combinational rule vector and deep reinforcement learning,” *Appl. Soft Comput.*, vol. 147, 2023, Art. no. 110802.
- [7] M. Ananthi and K. Vijayakumar, “Retracted article: stock market analysis using candlestick regression and market trend prediction (CKRM),” *J. Ambient Intell. Humanized Comput.*, vol. 12, no. 5, pp. 4819–4826, 2021.

- [8] LangChain, "Ai agents: Research & applications," Report 2024. Accessed: Apr. 2025.[Online]. Available: <https://accelxr.substack.com/p/ai-agents-research-and-applications>
- [9] X. Xu et al., "A survey on game playing agents and large models: methods, applications, and challenges," 2024, *arXiv:2403.10249*.
- [10] J. S. Park et al., "Generative agent simulations of 1,000 people," 2024, *arXiv:2411.10109*.
- [11] S. Ochoa et al., "Build a video search and summarization agent with NVIDIA AI blueprint," 2024. Accessed: Dec. 2024. [Online]. Available: <https://developer.nvidia.com/blog/build-a-video-search-and-summarization-agent-with-nvidia-ai-blueprint/>
- [12] X. Li and Y. Guo, "Paradigm shifts from data-intensive science to robot scientists," *Sci. Bull.*, vol. 70, no. 1, pp. 14–18, Jan. 2025.
- [13] L. Wu, S. Liu, and Y. Yang, "Grey double exponential smoothing model and its application on pig price forecasting in china," *Appl. Soft Comput.*, vol. 39, pp. 117–123, Feb. 2016.
- [14] S. Torbat, M. Khashei, and M. Bijari, "A hybrid probabilistic fuzzy arima model for consumption forecasting in commodity markets," *Econ. Anal. Policy*, vol. 58, pp. 22–31, Jun. 2018.
- [15] J. Li, G. Li, M. Liu, X. Zhu, and L. Wei, "A novel text-based framework for forecasting agricultural futures using massive online news headlines," *Int. J. Forecasting*, vol. 38, no. 1, pp. 35–50, 2022.
- [16] T. Xiong, C. Li, and Y. Bao, "Seasonal forecasting of agricultural commodity price using a hybrid STL and ELM method: Evidence from the vegetable market in China," *NeuroComputing*, vol. 275, pp. 2831–2844, Jan. 2018.
- [17] L. Zhang, J. Wang, and B. Wang, "Energy market prediction with novel long short-term memory network: Case study of energy futures index volatility," *Energy*, vol. 211, 2020, Art. no. 118634.
- [18] C.-C. Hung and Y.-J. Chen, "DPP: Deep predictor for price movement from candlestick charts," *PLoS One*, vol. 16, no. 6, 2021, Art. no. e0252404.
- [19] C.-C. Lin, Y.-C. Lai, M.-H. Zheng, C.-H. Wang, Y.-R. Chen, and L.-A. Gao, "The candlestick-tracking trend decision for day trading on Taiwan index futures market," in *Proc. 5th Int. Electron. Commun. Conf., Conf. Proc.*, 2023, pp. 69–77.
- [20] P. Gabrielsson and U. Johansson, "High-frequency equity index futures trading using recurrent reinforcement learning with candlesticks," in *Proc. IEEE Symp. Ser. Comput. Intell. IEEE, Conf. Proc.*, 2015, pp. 734–741.
- [21] D. Fengqian and L. Chao, "An adaptive financial trading system using deep reinforcement learning with candlestick decomposing features," *IEEE Access*, vol. 8, pp. 63666–63678, 2020.
- [22] W. J. O'Neil, *How to Make Money in Stocks*, McGraw-Hill: New York, NY, USA, 1988, vol. 10.
- [23] T. G. Fischer, "Reinforcement learning in financial markets-a survey," *FAU Discussion Papers in Econ.*, Rep. No.12/2018, 2018.
- [24] F. Rundo, "Deep LSTM with reinforcement learning layer for financial trend prediction in fx high frequency trading systems," *Appl. Sci.*, vol. 9, no. 20, p. 4460, 2019.
- [25] C. Ma, J. Zhang, J. Liu, L. Ji, and F. Gao, "A parallel multi-module deep reinforcement learning algorithm for stock trading," *NeuroComputing*, vol. 449, pp. 290–302, Aug. 2021.
- [26] E. Adamopoulou and L. Moussiades, "An overview of chatbot technology," in *Proc. IFIP Int. Conf. Artif. Intell. Appl. Innov. (AIAI), IFIP Adv. Inf. Commun. Technol.*, vol. 584, Cham, Switzerland: Springer, pp. 373–383, 2020.
- [27] R. L. Axtell and J. D. Farmer, "Agent-based modeling in economics and finance: Past, present, and future," *J. Econ. Literature*, vol. 63, no. 1, pp. 197–287, Mar. 2025.
- [28] A. Shavandi and M. Khedmati, "A multi-agent deep reinforcement learning framework for algorithmic trading in financial markets," *Expert Syst. Appl.*, vol. 208, 2022, Art. no. 118124.
- [29] D. Kanzari and Y. R. B. Said, "A complex adaptive agent modeling to predict the stock market prices," *Expert Syst. Appl.*, vol. 222, 2023, Art. no. 119783.
- [30] F.-F. He, C.-T. Chen, and S.-H. Huang, "A multi-agent virtual market model for generalization in reinforcement learning based trading strategies," *J. Econ. Literature*, vol. 134, 2023, Art. no. 109985.
- [31] M. E. Pérez-Pons, R. S. Alonso, O. García, G. Marreiros, and J. M. Corchado, "Deep q-learning and preference based multi-agent system for sustainable agricultural market," *Sensors*, vol. 21, no. 16, p. 5276, 2021.
- [32] OpenAI, "Introducing chatGPT," 2022. Accessed: Nov. 2024. [Online]. Available: <https://openai.com/blog/chatgpt>
- [33] J. S. Park, J. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative agents: Interactive simulacra of human

behaviour," in *Proc. 36th Annu. ACM Symp. User Interface Softw. Technol.*, pp. 1–22, 2023.

- [34] C. Li and D. J. Hayes, "Price discovery on the international soybean futures markets: A threshold co-integration approach," *J. Futures Markets*, vol. 37, no. 1, pp. 52–70, 2017.
- [35] M. T. Bohl, P. L. Siklos, and C. Wellenreuther, "Speculative activity and returns volatility of Chinese agricultural commodity futures," *J. Futures Markets*, vol. 54, pp. 69–91, Feb. 2018.
- [36] B. Shiferaw, M. Smale, H.-J. Braun, E. Duveiller, M. Reynolds, and G. Muricho, "Crops that feed the world 10. Past successes and future challenges to the role played by wheat in global food security," *Food Secur.*, vol. 5, pp. 291–317, 2013.



Jiarong Wang (Graduate Student Member, IEEE) received the B.S. degree in automation from Changchun University of Science and Technology, Changchun, China, in 2019. She is currently working toward the Ph.D. degree in electronic science and technology with the School of Electronic Engineering, Xidian University, Xi'an, China.

Her research interests include AI Agents and memory management in chatbots.



Jiaji Wu (Member, IEEE) received the B.S. degree in electrical engineering from Xidian University, Xi'an, China, in 1996, the M.S. degree in astrometry and celestial mechanics from the National Time Service Center, Chinese Academy of Sciences, Xi'an, in 2002, and the Ph.D. degree in electrical engineering from Xidian University, in 2005.

Currently, he is a Professor with Xidian University. His research interests include still image coding, hyperspectral/multispectral image processing, communication, big data, the Internet of Things (IoT), and high-performance computing.



Guoshuai Zhang received the B.S. degree in communication engineering from Changchun University, Changchun, China, in 2016, the M.S. degree in signal and information processing from the South-Central Minzu University, Wuhan, China, in 2019, and the Ph.D. degree in electronics science and technology from Xidian University, Xi'an, China, in 2023.

Currently, he is a Postdoctoral Researcher with Xidian University. His research interests include virtual game and social networks.



Mingzhou Tan received the B.S. degree in electrical engineering from Xidian University, Xi'an, China, in 1996.

Currently, he is an Adjunct Professor with Xidian University; a Partner of AAMA Angel Investment Management (Beijing) Company Ltd., Beijing, China; and a Visiting Professor and an Entrepreneurship Mentor of many famous universities. His research interests include digital economy, artificial intelligence, big data, and metaverse.



Shaohong Chen received the B.S. degree in intelligence science and technology, the M.S. degree in electronic information from Xidian University, Xi'an, China, in 2020 and 2025, respectively. He is currently working toward the Ph.D. degree in electronic science and technology with the School of Electronic Engineering, Xidian University.

His research interests include AI art creation and intelligent video creation.

Zeyi Lin received the B.S. degree in intelligence science and technology from Xidian University, Xian, China, in 2022. He is currently enrolled in a combined M.S.–Ph.D. program in electronic science and technology with the School of Electronic Engineering, Xidian University.

His research interest includes AI art creation.

Zeyi Lin, photograph not available at the time of publication.